

THAARU Sepsis AI — Clinical Preprocessing

Data Engineering & Preprocessing Pipeline Summary • Generated: July 19, 2026

1. Preprocessing Executive Summary

This report documents the clinical range cleaning, time-series imputation, outlier handling, and standard scaling executed on the consolidated PhysioNet Sepsis dataset. To ensure research rigor and prevent data leakage, imputation medians and scaling parameters were fit exclusively on the Training set cohort and applied symmetrically across all partitions. The processed cohort contains 66 clinical features after appending missingness indicator features.

Pipeline Step	Strategy Employed
Patient Split	70% Train, 15% Val, 15% Test (Patient-wise Stratified)
Range Validation	Physiological bounds checking; Out-of-bound replaced with NaN
Vitals Imputation	Patient-wise forward-fill (limit=6h) + Train Medians
Labs Imputation	Patient-wise forward-fill (unlimited) + Train Medians
Indicators	26 binary flags tracking lab measurement ordering
Outlier Handling	0.1% - 99.9% clipping fit on Train set to remove extreme noise
Feature Scaling	StandardScaler fit on Train set and applied to all splits

2. Processed Dataset Partition Dimensions

Split	Patients Count	Total Hour Records	Sepsis Prevalence (%)
Training Split	28,235	1,086,436	7.27%
Validation Split	6,050	231,472	7.27%
Testing Split	6,051	234,302	7.27%

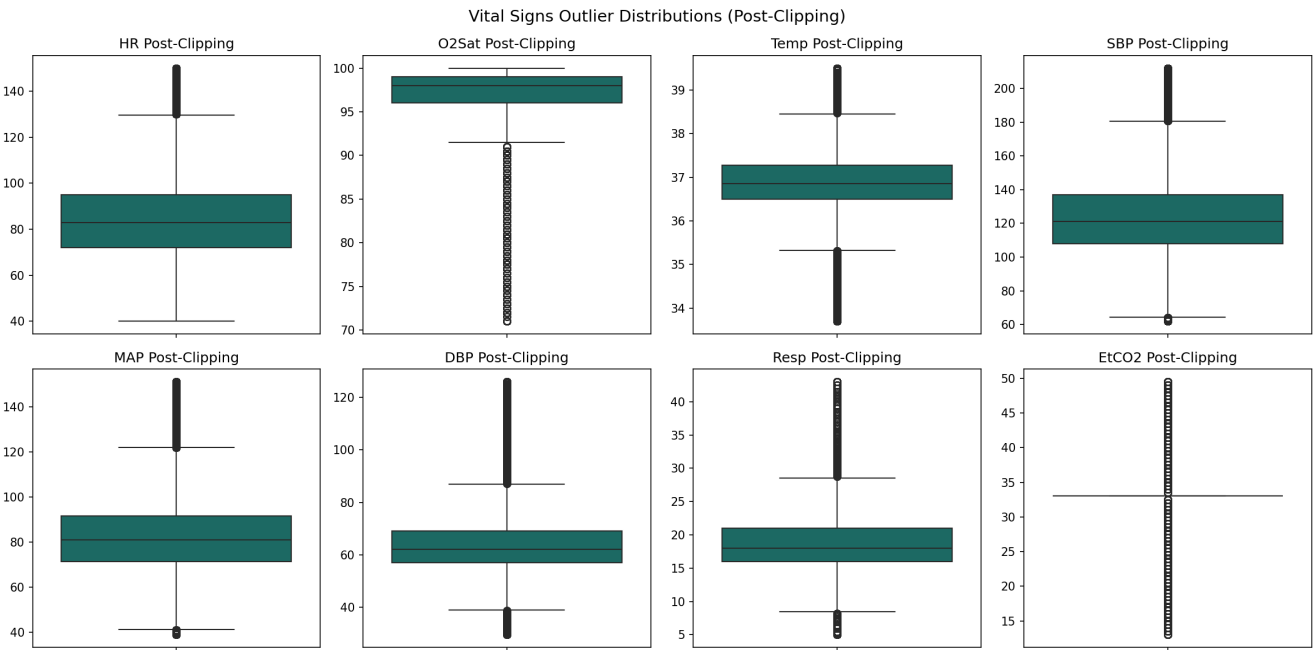
3. Clinical Range Validation Summary

Out-of-bound physiological parameters represent artifactual anomalies (e.g. lead disconnects or clinical entry errors). These records were replaced with NaN. The table below summaries the replaced count per feature:

Feature	Physiological Limit	Train Replacements	Val Replacements	Test Replacements
HR	20 - 250	0	4	0
Temp	25 - 45	5	0	1
Resp	4 - 80	1,333	288	216
SBP	40 - 300	63	27	24
MAP	20 - 200	286	85	73
DBP	20 - 200	55	20	28
pH	6.8 - 8.0	8	1	2

4. Post-Clipping Outlier Visualizations

To preserve real medical critical states while capping sensor extremes, vital signs were clipped at the conservative 0.1% and 99.9% percentiles computed on the training partition.



5. Quality Assurance Checklist

A critical validation pass was executed on the processed splits. All verification steps passed successfully, confirming split isolation and data completeness.

Quality Check Criteria	Status	Details / Verification Counts
Missing Values Check	PASSED	0 missing values in train, val, and test splits
Duplicate Rows Check	PASSED	0 duplicate records in all split files
Patient Leakage Check	PASSED	Mutual overlaps: Train/Val: 0, Train/Test: 0
Split Feature Count Alignment	PASSED	Symmetric schemas with exactly 68 features

6. Preprocessing Conclusions & Guidelines

By enforcing patient-wise stratified partition, adding lab missingness indicators, capping statistical extremes, and scaling features via Train set mean and variances, we have established a leak-proof, reproducible data preprocessing foundation.

This dataset is ready for Phase 4 sequence generation and baseline/LSTM model training.